UNIT 7: DIFFERENTIAL EQUATIONS

First off, we shall address the question: *what is a differential equation?* For the answer to this question, we may need to look back at ordinary equations, those with terms that are each some expression of *x*, for some sort of conceptual reference. consider

$$x^5 + 6x^3 + x = 9.$$

In the above equation, we are to solve for x, and in such an equation, there may be multiple solutions. Differential equations aren't so different than equations like the above; the only difference is that differential equations are in terms of *y*, or *f (x)*, and instead of *y* being raised to exponents in the way *x* is in the above example, *y* is differentiated with respect to *x* to varying orders. Letting *y = y (x)*, consider:

$$y'' + 5y' + y = 6.$$

In one term, there is a second derivative, in another, there is a first derivative, there's also a term with the function itself, and all of these varying orders of differentiation of *y* make an equation. Our goal in solving differential equations is to solve for *y*. And like in the case of the ordinary equation, we may get a variety of solutions, depending on the differential equation.

Ex 1. Solve the following differential equation.

$$\mathbf{y' - y = 0}$$

The first thing that may be convenient to note is that we can add y to both sides such that

$$y' = y.$$

Now, this equation is telling us that whatever y is, it's the derivative of itself, meaning that if we differentiate y with respect to x, we get the function we started with. One such function is e^x *, where*

$$\frac{d}{dx}e^x = e^x.$$

This means that e^x *is a solution. But, the more general solution for y is*

$$ne^x \quad \textbf{\textit{*for all real numbers n}}$$

since

$$\frac{d}{dx}ne^x = n\frac{d}{dx}e^x = ne^x.$$

In **Ex 1**, we experienced a simple **first-order differential equation**. That means that the highest order of differentiation is 1, where there's only a first derivative - not a second, third, or even fourth derivative. The first differential equation we showed as a preview has a second derivative, so it's a **second-order differential equation**.

Sometimes in a differential equation, a derivative of *y* (of any order) can be related to an expression of x. Consider the following example.

Ex 2. Solve the following differential equation.

$$y'' = 3x^5$$

There is only one term in which y appears, and it's under multiple layers of differentiation. In this kind of differential equation, y is trapped within an "onion", and we get to y by peeling off the layers that enclose it. Here, the layers are derivatives, so we peel them off by integrating both sides of the equation - as many times as necessary. y is trapped under two layers of differentiation, so we need two layers of integration. Integrating both sides of the equation twice gets us $y = \frac{x^7}{14} - Cx$.

$$\int \left(\int y'' \, dx\right) dx = \int \left(\int 3x^5 \, dx\right) dx$$

$$\int (y' + C) \, dx = \int \frac{x^6}{2} dx$$

$$y + Cx = \frac{x^7}{14}$$

$$y = \frac{x^7}{14} - Cx.$$

Typically, differential equations also come with **initial conditions**. In other words, we may have to solve the differential equation in **Ex 2** with extra information given to us at the beginning. Usually, this extra information is given in the form

y({some given input}) = {some given output}.

Consider the following example, which is a slight variation of **Ex 2**.

Ex 3. Given the initial condition *y (1) = 1*, solve the differential equation from Ex 2.

*From **Ex 2**, we know*

$$y(x) = \frac{x^7}{14} - Cx.$$

The initial condition allows us to solve for the ambiguous constant C,

$$y(1) = \frac{(1)^7}{14} - C = 1$$

$$C = \frac{(1)^7}{14} - 1 = -\frac{13}{14}$$

Now we substitute the C value into our general solution to get the more specific solution

$$y(x) = \frac{x^7}{14} + \frac{13}{14}x.$$

As seen in **Ex 3**, initial conditions restrict y to a smaller solution set; they eliminate ambiguous constants that allow differential equations the wiggle room to have infinitely many solutions.

Separable First-order Differential Equations

In this unit, we will be focusing primarily on first-order differential equations, of which there are two main types: **separable** first-order differential equations and **inseparable** differential equations. Separable differential equations follow the form

$$\frac{dy}{dx} = f(x)\, g(y)$$

where the first derivative of y can be expressed as a product of a function of y (namely, $g(y)$) and a function of x (namely, $f(x)$). We call them *separable* since we can take the general form and rewrite it as

$$\frac{dy}{g(y)} = f(x)\, dx$$

where we are taking everything x-specific and moving it to one side of the equation and moving everything y-specific to the other side of the equation. The x-stuff and y-stuff have been organized in such a manner that each side is a product between a differential (we call dx and dy differentials) and either a function or the reciprocal of a function. We want to do this since it so conveniently sets us up to integrate both sides; since the left side is equal to the right side of the above equation, an infinite sum of the left side is the same thing as an infinite sum of the right side (and infinite sums of *something times a differential* gets you an integral). Taking advantage of this convenient setup, we get

$$\int \frac{dy}{g(y)} = \int f(x)\, dx\,.$$

With the first-order differential equations that we call inseparable, a first derivative of y is equal to some expression of y and x that cannot be expressed as $f(x)\, g(y)$, which takes away the option of making each side a product between a function and a differential. We cannot *separate* the x-stuff and y-stuff to conveniently set up two simple integrals; this is of course, why we call such equations *inseparable*.

Ex 4. Solve.

$$\mathbf{y' = 8x^5y^3}$$

By looking at this differential equation, we can notice how everything is associated by multiplication, which makes it convenient to say that since $y' = \frac{dy}{dx}$,

$$\frac{dy}{y^3} = 8x^5\, dx\,,$$

which beautifully sets us up to solve both sides of

$$\int y^{-3} dy = \int 8x^5\, dx$$

in order to get

$$-\frac{y^{-2}}{2} + C = 8\left(\frac{x^6}{6} + W\right) \qquad \text{*}W \text{ is another ambiguous constant}$$

$$-\frac{1}{2y^2} + C = 8(\frac{x^6}{6} + W)$$

from which we solve for y, which gets us $y = \sqrt{\frac{1}{-8\frac{x^6}{3}+C}}$

$$-\frac{1}{2y^2} = 8(\frac{x^6}{6} + W) - C$$
$$\frac{1}{2y^2} = -4\frac{x^6}{3} - 8W + C$$
$$\frac{1}{y^2} = -8\frac{x^6}{3} - 2(8W - C)$$

**we don't care about* $-2(8W - C)$ *since it's just another ambiguous constant, so we just call it C*

$$\frac{1}{y^2} = -8\frac{x^6}{3} + C$$
$$y = \sqrt{\frac{1}{-8\frac{x^6}{3}+C}}$$

Slope Fields

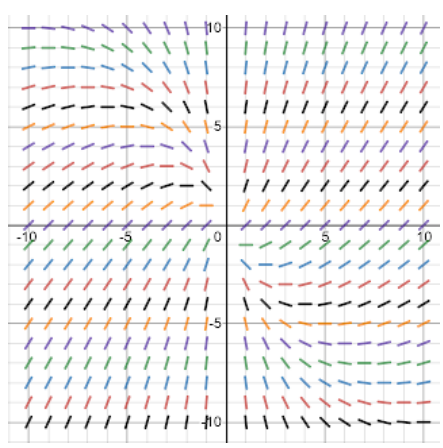
We can also solve differential equations using **slope fields**. Consider the following differential equation.

$$\frac{dy}{dx} = \frac{x+y}{x}$$

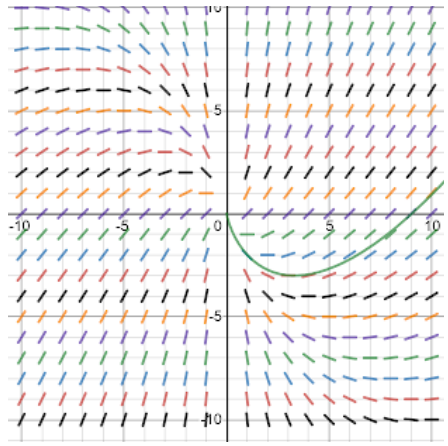
The first derivative of *y(x)* cannot easily be expressed as the product of a function of *y* and a function of *x*, making it an inseparable differential equation; thus, the method used for separable first-order differential equations isn't as available as in **Ex 4**. However, it comes to our convenience to notice how the first derivative is defined in terms of *x* and *y*, making it vary in value over different points on the x-y plane, as we plug in the *x* and *y* values from the x-y coordinates.

$$\frac{dy}{dx} = \frac{x+y}{x} = S(x,y)$$

Points on the x-y plane spit out slope values - those of *y(x)*. When, for each point, one attaches an intersecting line with a slope equal to *S(x,y)*, the result is an x-y plane full of lines, which is what we call a slope field. Here's the slope field for the above differential equation.



Notice how we can create curves by connecting certain lines. Since these lines are tangent to *y(x)*, the curves we create by connecting them are the solutions to our differential equation.



This methodology is more geometric and approximate, getting us to only a *picture* of the solutions and not directly to the general analytic solution itself,

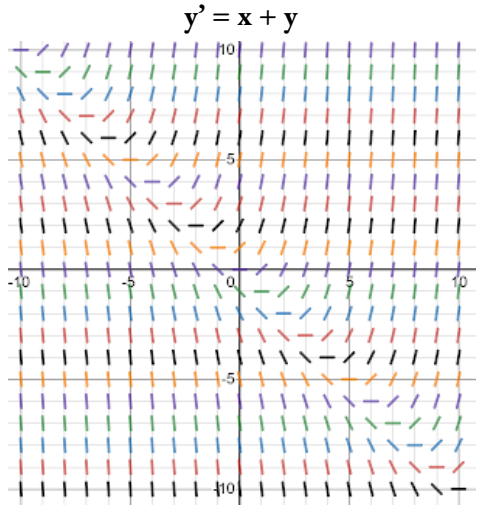
$$y = xe^{C}(\ln x + D) \qquad \text{*define } C \text{ and } D \text{ as ambiguous constants}$$

which can be obtained via a more advanced method of solving differential equations that we will not be teaching.

The key, when using slope fields, is to be able to recognize functions from curves. One should be able to look at some curve and think "Oh. That looks like a logarithmic function," or "Hm. This might be a cosine function. It crosses through the origin? Never mind, that's a sine function!" Consider the following example.

Ex 5. Which of the following functions is a solution to the given differential equation?

y' = x + y



$g(x) = ln(x)$ $\quad$ $h(x) = e^x - x - 1$ $\quad$ $w(x) = cos(x) + sin(x)$ $\quad$ $r(x) = \frac{2x+7}{x^3+1}$

As a logarithmic function approaches increasing x-values, the curve flattens out, which is not inherent in the behavior of the slope field. g(x) cannot be a solution.

w(x) has no defined limit as x goes to infinity, whereas in the slope field, there is a common trend that suggests that as x approaches infinity, so does every solution; hence, w(x) is not a solution.

r(x) is another one of those functions that flattens out as x approaches infinity, so it can't be a solution.

We are left with h(x), which seems to fit the behavior of the slope field; as x approaches negative-infinity, the first term approaches zero and the remaining term goes to infinity. This fits the behavior of our slope field. Also, since the first term goes to zero, as we move farther and farther backward in the x-y plane, the function should look increasingly like y = -x - 1; if we look carefully at the straight line solution on the slope field, notice that it is exactly y = -x - 1, and all curves on the slope field seem to converge to this line, so h(x) fits the common behavior of the lines on the slope field. Lastly, note how all lines on the positive-x side seem to fit the behavior of something with an exponential function in it, this very description matches our candidate. Out of all of the given functions, h(x) has to be a solution to the differential equation.

Euler's Method

When all else, as far as methodology, can't be done, another method for solving first-order differential equations is **Euler's method**, a numerical approximation method. Euler's method helps us approximate the value of the solution function at a given x-value. If we use Euler's method on an interval within the x-axis, we recreate an approximation to a solution curve as defined in that interval.

Typically, we start with a differential equation and an initial condition $y(x_0) = y_0$. We then plug x_0 and y_0 into the differential equation to obtain the value of the first derivative, or $S(x,y)$. We then jump forward in the x-axis some Δx and the derivative of the solution function gives us a hint as to how far from the y-value in the initial condition we have to jump in order to obtain the next y-value.

Consider the limit-definition of the derivative covered in an earlier unit:

$$y'(x) = \lim_{\Delta x \to 0} \frac{y(x+\Delta x) - y(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

The above definition is a limit as $\Delta x \to 0$ of an approximation of $y'(x)$; the difference quotient is an approximation that we can refine by assigning the x-increment smaller and smaller values. Therefore, we can say

$$y'(x) \approx \frac{y(x+\Delta x) - y(x)}{\Delta x} = \frac{\Delta y}{\Delta x},$$

and by extension,

$$y'(x)\Delta x \approx y(x+\Delta x) - y(x) = \Delta y.$$

But, considering that we are making an approximate solution curve, let's assume that both sides are equal.

$$y'(x)\Delta x = y(x+\Delta x) - y(x).$$

If we plug in the x-value we jumped to after the first x-increment, we get

$$y'(x_0 + \Delta x)\,\Delta x = y(x_0 + 2\,\Delta x) - y(x_0 + \Delta x) = \Delta y$$
$$y'(x_0 + \Delta x)\,\Delta x = \Delta y\,.$$

We then add this y-increment to the y-value in the initial condition to get the next y-value. After we repeat this process *n* times, we hop *n* x-increments, hence the following:

$$y'(x_0 + n\Delta x)\,\Delta x = \Delta y\,.$$

The y-increments change as we move along the x-axis; the *i*th y-increment is

$$y'(x_0 + i\Delta x)\,\Delta x = \Delta y_{i-1}\,.$$

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Another way to derive a formula for the *i*-th y-increment is the following: we can define the slope of a secant line on the solution curve as *m*. Since

$$m = \frac{\Delta y}{\Delta x}\,,$$

we can solve for the y-increment by multiplying both sides by the x-increment, and we can plug in the first derivative of the solution function into *m* - since a derivative is a slope (and since the mean value theorem tells us that there has to be some x-value *c* within the Δx interval for which $y'(c)$ is equal to the slope of the secant line, making *m* an approximation of y'(x) in that interval). To get the next y-value, we add the y-increment to the initial y-value. Call this y_1.

$$m\Delta x = \Delta y$$
$$y'(x_0)\,\Delta x = \Delta y$$
$$y'(x_0)\,\Delta x + y_0 = y_1$$

We can iterate this process, hopping another Δx on the x-axis, in order to obtain y_2.

$$y'(x_1)\,\Delta x + y_1 = y_2$$

Define the x-value we stop at to be x_1, where

$$x_0 + \Delta x = x_1\,.$$

Hence,

$$y'(x_0 + \Delta x)\Delta x + y_1 = y_2.$$

After another iteration, we get y_3,

$$y'(x_0 + 2\Delta x)\Delta x + y_2 = y_3.$$

If we look carefully at the expressions for y_1, y_2, and y_3, we'll notice a general relationship between one y-value and another that follows it:

$$y'(x_0 + i\Delta x)\Delta x + y_i = y_{i+1}.$$

If we subtract y_i from y_{i+1}, we get the i-th y-increment; we have observed the y-increments to depend on the number of iterations.

$$y'(x_0 + i\Delta x)\Delta x = \Delta y_{i-1}.$$

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We can construct a formula that gives us the y-value after n iterations - namely, a formula for y_n:

$$y_0 + \sum_{i=0}^{n} \Delta y_i = y_n$$

$$y_0 + \sum_{i=1}^{n} y'(x_0 + i\Delta x)\Delta x = y_n.$$

After one iteration,

$$y_0 + y'(x_0 + \Delta x)\Delta x = y_1,$$

which is true.

Ex 6. Estimate *y(0.3)* given *y(0) = 1* and the following first-order differential equation.

$$\mathbf{y' = -\frac{x}{y}.}$$

Let's make the x-increment very small in comparison to the 0.3 for the sake of precision; let our Δx be equal to 0.03. Respectively, After 10 iterations, we will get to x = 0.3 and a corresponding $y_{10} \approx y(0.3)$. Given the available information,

$$1 + \sum_{i=1}^{10} y'(i\Delta x)\Delta x = y_{10}$$

$$1 - \sum_{i=1}^{10} i\,(0.03)^2 = y_{10}$$

$$0.9505 = y_{10}$$

Thus we obtain our answer:

$$y(0.3) \approx 0.9505.$$

Exercises

1. $y' = x + y^2$
2. $y' = 1 + y^2,\ y(0) = 0$
3. $(x^2 + 1)y' = (xy)^2$ **,** $y(1) = 2$
4. $y' = \frac{3}{x}$ **, approximate the y-values with 3 iterations of** $\Delta x = 0.5$
5. $y' = y$